# Logical Relational Model

Library Management System — EECS447: Database Systems
Fall 2026 · Professor Hossein Saiedian
Team Name: DataBASED

## 1. Introduction

Project Overview:

> This database will be a reliable library management system that efficiently manages library books, magazines, digital media, client memberships, borrowing and returning transactions, reservations, and fees. The system will help library staff manage library operations and allow clients to search for items, reserve books, and check their loan status on their library membership. The database will also provide useful reports for both users for library fines, book availability, client activity, and more.

Scope:

> The library management system will manage books, digital media, and magazines; client accounts and membership types; borrowing and return transactions; reservations; and notifications for due dates, overdue items, and available reservations. Membership types will determine borrowing limits and fee rules. The system will generate reports on fines, item availability, borrowing trends, and client activity. It will include a staff interface for managing items, loans, returns, and accounts, and a client interface for searching the catalog, reserving items, and checking loan status. Work outside these library collection, circulation, and account functions is beyond the scope of this project.

## 2. Relational Schema Mapping

- Identify relations

| ER component | Type | Relation |
|---|---|---|
| Item (superclass) | Entity | ITEM |
| Book | Entity (subclass of Item) | BOOK |
| Digital Media | Entity (subclass of Item) | DIGITAL MEDIA |
| Magazine | Entity (subclass of Item) | MAGAZINE |
| Membership | Entity | MEMBERSHIP |
| User (Client) | Entity | CLIENT |
| Loan | Relational entity | LOAN |
| Reservation | Relational entity | RESERVATION |
| ISA | Specialization | None; each subclass share the |

| | | |
|---|---|---|
| | | PK,FK itemID from ITEM |
| Has (Membership to Client) | Relationship | None; MembershipType is a FK in CLIENT |
| Has (Loan to Item) | Relationship | None; itemID is a FK in Loan |
| Has (Reservation to Item) | Relationship | None; itemID is a FK in Reservation |
| Makes (Client to Loan) | Relationship | None; ClientID is a FK in Loan |
| Makes (Client to Reservation) | Relationship | None; ClientID is a FK in Reservation |

- Define attributes and domains

ITEM

| Attributes | Type | Domain/Constraints | Description |
|---|---|---|---|
| ItemID | INT | PK, positive, and incremental integer | Unique identifier assigned to each item |
| Title | VARCHAR(255) | Not `NULL` | Title of item |
| AvailabilityStatus | VARCHAR(20) | Not `NULL`; ‘Available’ or ‘Checked out’ | Whether the item can currently be borrowed |
| Restricted | VARCHAR(20) | Not `NULL`; ‘None’ or ‘Restricted’ | Whether borrowing of the item is restricted |

BOOK

| Attributes | Type | Constraints | Description |
|---|---|---|---|
| ItemID | INT | PK, FK → ITEM(ItemID) | Unique identifier assigned to each item; References the PK from ITEM |
| Author | VARCHAR(100) | Not `NULL` | Author of the book |
| Genre | VARCHAR(50) | Not `NULL` | Genre of the item |
| PubYear | YEAR | N/A | Year the book was |

| | | | |
|---|---|---|---|
| | | | published |
| ISBN | CHAR(13) | Not ‘NULL’, Unique | International Standard Book Number |
| Format | VARCHAR(30) | N/A | Format of the book |
| Publisher | VARCHAR(20) | N/A | Company that published the book |

DIGITAL MEDIA

| Attributes | Type | Constraints | Description |
|---|---|---|---|
| ItemID | INT | PK,<br>FK → ITEM(ItemID) | Unique identifier assigned to each item; References the PK from ITEM |
| Creator | VARCHAR(100) | Not `NULL` | Creator of the media |
| Format | VARCHAR(30) | N/A | Format of item (movie, music, tv show etc.) |
| Genre | VARCHAR(50) | Not `NULL` | Genre of the item |
| ReleaseYear | YEAR | N/A | Year item was released |

MAGAZINE

| Attributes | Type | Constraints | Description |
|---|---|---|---|
| ItemID | INT | PK,<br>FK → ITEM(ItemID) | Unique identifier assigned to each item; References the PK from ITEM |
| IssueNum | VARCHAR(20) | N/A | Issue number of magazine |
| PubDate | DATE | N/A | Date magazine was published |
| Publisher | VARCHAR(20) | N/A | Company that published the magazine |

MEMBERSHIP

| Attributes | Type | Constraints | Description |
|---|---|---|---|
| MembershipType | VARCHAR(10) | Unique and must be ‘Regular’, ‘Student’, or ‘Seniors’ | Type of memberships available |
| BorrowingLimit | INT | Not ‘Null’ | Max amount of items member can borrow |
| DailyFine | DECIMAL(5,2) | Between 3 and 8 | Amount of money charged for late returns |
| LoanPeriod | INT | Not ‘NULL’ | Amount of days member can borrow items for |
| ReserveLimit | INT | Not ‘NULL’ | Max amount of items member can reserve |

CLIENT

| Attributes | Type | Constraints | Description |
|---|---|---|---|
| ClientID | INT | Positive, unique, and incremental integer | Unique client identifier |
| FirstName | VARCHAR(50) | Not ‘NULL’ | First name of client |
| LastName | VARCHAR(50) | Not ‘NULL’ | Last name of client |
| Email | VARCHAR(250) | N/A | Email address of client |
| AccountStatus | VARCHAR(20) | “Active” or “Suspended” | Status of client’s account |
| FeesOwed | DECIMAL(5,2) | N/A | Total amount of fees owed by client |
| MemberType | VARCHAR(10) | FK → MEMBERSHIP(MembershipType) | Membership type of client |

LOAN

| Attributes | Type | Constraints | Description |
|---|---|---|---|
| LoanID | INT | PK, positive, and incremental integer | Identifier of loan |

| | | | |
|---|---|---|---|
| ClientID | INT | FK → CLIENT(ClientID), Not NULL | Identifier of client |
| ItemID | INT | FK → ITEM(ItemID), Not NULL | Identifier of Item |
| CheckoutDate | DATE | Not 'NULL' | Date items checked out |
| DueDate | DATE | Not 'NULL' | Date items due back |
| ReturnDate | DATE | N/A | Date items returned by client |
| FineAmount | FLOAT | N/A | Late fees accrued from loan |

RESERVATION

| Attributes | Type | Constraints | Description |
|---|---|---|---|
| ReservationID | FLOAT | PK, positive, and incremental integer | Identifier of reservation |
| ClientID | FLOAT | FK → CLIENT(ClientID), Not NULL | Identifier of client |
| ItemID | FLOAT | FK → ITEM(ItemID), Not NULL | Identifier of item |
| ReservationDate | DATE | Not 'Null' | Date reservation was made |
| ReservationStatus | VARCHAR(20) | Not NULL; "Pending", "Ready", "Expired", or "Fulfilled" | Status of reserved item |

- Determine primary keys

| Relation | Primary Key |
|---|---|
| Item | ItemID |
| Book | ItemID |
| Digital Media | ItemID |
| Magazine | ItemID |

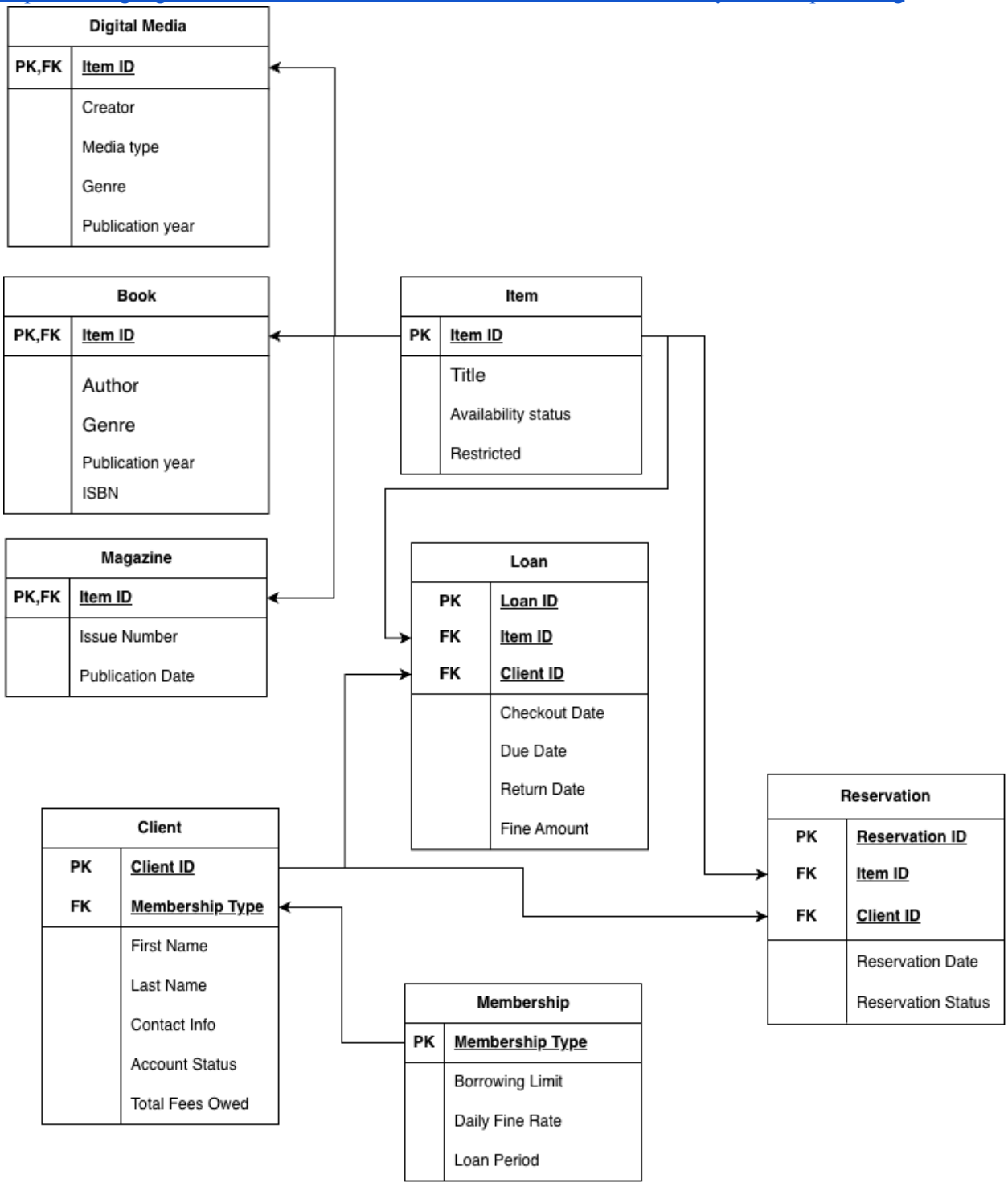
| | |
|---|---|
| Membership | MembershipType |
| Client | ClientID |
| Loan | LoanID |
| Reservation | ReservationID |

- Establish foreign keys

| Relation | Foreign Key | References |
|---|---|---|
| Book | ItemID | Item(ItemID) |
| Digital Media | ItemID | Item(ItemID) |
| Magazine | ItemID | Item(ItemID) |
| Client | MembershipType | Membership(MembershipType) |
| Loan | ClientID | Client(ClientID) |
| Loan | ItemID | Item(ItemID) |
| Reservation | ClientID | Client(ClientID) |
| Reservation | ItemID | Item(ItemID) |

3. Relational schema diagram

https://drive.google.com/drive/folders/1chZW5eF8LOi55M-m4x8tVznibE8yhCb1?usp=sharing



## 4. Schema Documentation

| Attributes | Type | Domain/Constraints | Description |
|---|---|---|---|
| ItemID | INT | PK, positive, and incremental integer | Unique identifier assigned to each item |
| Title | VARCHAR(255) | Not `NULL` | Title of item |
| AvailabilityStatus | VARCHAR(20) | Not `NULL`; ‘Available’ or ‘Checked out’ | Whether the item can currently be borrowed |
| Restricted | VARCHAR(20) | Not `NULL`; ‘None’ or ‘Restricted’ | Whether borrowing of the item is restricted |
| Author | VARCHAR(100) | Not `NULL` | Author of the book |
| Genre | VARCHAR(50) | Not `NULL` | Genre of the item |
| PubYear | YEAR | N/A | Year the book was published |
| ISBN | CHAR(13) | Not ‘NULL’, Unique | International Standard Book Number |
| Format | VARCHAR(30) | N/A | Format of the book |
| Publisher | VARCHAR(20) | N/A | Company that published the book |
| Creator | VARCHAR(100) | Not `NULL` | Creator of the media |
| Genre | VARCHAR(50) | Not `NULL` | Genre of the item |
| ReleaseYear | YEAR | N/A | Year item was released |
| IssueNum | VARCHAR(20) | N/A | Issue number of magazine |
| PubDate | DATE | N/A | Date magazine was published |
| MembershipType | VARCHAR(10) | Unique and must be ‘Regular’, ‘Student’, or ‘Seniors’ | Type of memberships available |
| BorrowingLimit | INT | Not ‘Null’ | Max amount of items member can borrow |

| | | | |
|---|---|---|---|
| DailyFine | DECIMAL(5,2) | Between 3 and 8 | Amount of money charged for late returns |
| LoanPeriod | INT | Not ‘NULL’ | Amount of days member can borrow items for |
| ReserveLimit | INT | Not ‘NULL’ | Max amount of items member can reserve |
| ClientID | INT | Positive, unique, and incremental integer | Unique client identifier |
| FirstName | VARCHAR(50) | Not ‘NULL’ | First name of client |
| LastName | VARCHAR(50) | Not ‘NULL’ | Last name of client |
| Email | VARCHAR(250) | N/A | Email address of client |
| AccountStatus | VARCHAR(20) | “Active” or “Suspended” | Status of client’s account |
| FeesOwed | DECIMAL(5,2) | N/A | Total amount of fees owed by client |
| CheckoutDate | DATE | Not ‘NULL’ | Date items checked out |
| DueDate | DATE | Not ‘NULL’ | Date items due back |
| ReturnDate | DATE | N/A | Date items returned by client |
| FineAmount | FLOAT | N/A | Late fees accrued from loan |
| ReservationDate | DATE | Not ‘Null’ | Date reservation was made |
| ReservationStatus | VARCHAR(20) | Not NULL; “Pending”, “Ready”, “Expired”, or “Fulfilled” | Status of reserved item |